Sungjun Heo

Ulsan, Republic of Korea | Personal Website | Lab Website | Sungjun Heo

Education

Integrated M.S.–Ph.D. — **Ulsan National Institute of Science and Technology (UNIST)** — Mar 2024 -
- Graduate School of Artificial Intelligence
- Robotics & Mobility Laboratory of Prof.J Jeon
- GPA : 4.0 / 4.3

B.S — **Ulsan National Institute of Science and Technology (UNIST)** — Mar 2021 - Fed 2024
- Department of Electrical Engineering
- Early Graduation (by one year)
- GPA : 3.6 / 4.3 , Total credits : 127

HS — **Changwon Science High School (CSHS)** — 2018 - 2020

Publications

Conference Publications

[C1] **SHIFT-RL: Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments**
Sungjun Heo and Jeong hwan Jeon
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

[C2] **Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving**
Sungjun Heo, Heedon Jeong, Eunchong Kim, Hyojae Lee, Jungeun Lee, Seongjae Lee, et al.
IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026. **(Oral Presentation)**

[C3] **Decoupling Actor and Critic Replay for Offline Reinforcement Learning with Three-Factor Selection**
Sungjun Heo and Jeong hwan Jeon
IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026. **(Oral Presentation)**

[C4] **Deployment-Oriented End-to-End Autonomous Driving: Enhancing Closed-Loop Stability with a Lightweight Framework**
Eunchong Kim, Heedon Jeong, **Sungjun Heo**, Sunhwi Kim, Seongjae Lee, Jaechan Shin, Heecheol Yoo, Jeong hwan Jeon
IEEE Intelligent Vehicles Symposium (IV), 2026, pp. 694–701. **(Oral Presentation)**

[C5] **Predicted State-Based Hierarchical Reinforcement Learning for Long-Term Decision Making in Urban Dynamic Scenarios**
Sungjun Heo and Jeong hwan Jeon
IEEE Intelligent Vehicles Symposium (IV), 2025, pp. 1164–1170.

Preprint / Arxiv / Under Review

[A1] **MoE-DAgger: Transition-Aware Router-Entropy Gating for Selective Human Correction in Phase-Structured Manipulation**
Sungjun Heo, Sungjun Yang, Heedon Jeong, Junhyoung Ha, Jeong hwan Jeon

Research Interest

Autonomous Driving (AD)
- End-to-End (E2E) driving, Trajectory planning, Spatiotemporal fusion model

Robot manipulation
- Learning based manipulation, Humanoid motion tracking, Multi-Agent

Reinforcement Learning (RL)
- Hierarchical, Curriculum reinforcement learning

Awards and Achievements

First Prize | AI Technology Open Workshop Oct 2026

- AI Top Conference Poster session (ICML, CVPR, ECCV, IJCAI, IROS, ICLR, ACL, KDD, AAAI„)
- Presentation : SHIFT-RL: Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments (2026 IROS)

IEEE SMC 2026 Student and Young Professional Travel Grant Oct 2026
- Awarded $600

- IEEE Systems, Man, and Cybernetics Society

Grand Prize | Brain To Society, U-Challenge Festival (1 / 96 teams) Dec 2025
- Awarded $3,600

- Brain To Society : Future Mobility part
- Project Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments (2026 IROS)

Excellence Award | AI Technology Open Workshop Oct 2025
- Awarded $2,400

- UNIST AI Tech Open Workshop | Team FAST
- Research title : Development of an End-to-End Autonomous Driving Model Based on a High-Dimensional Simulator (2026 IV)

1st place | 2025 Hyundai Motor Group's Autonomous Driving Challenge (1 / 16 teams) Sep 2025
- Awarded $22,000 + (Special privilege) Autonomous driving field trip

- Final 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Full-Self-Driving in K-City

3rd place | Preliminary Round (3 / 16 teams) Mar 2025
- Awarded $3,600

- Preliminary Round of 2025 Hyundai Motor Group's Autonomous Driving Challenge
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

Undergraduate Research Award (Top 5 / All graduation students) Dec 2023

- Research Title : Autonomous Driving Waypoint-generation based on Vehicle Kinematics.

Academic Achievement Award 2021

- Undergraduate academic achievement excellence award in 2021

Work Experience

Paper Reviewer

- 3 Papers in 2025 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 2 Papers in 2025 IEEE International Conference on Intelligent Transportation Systems
- 2 Papers in 2026 IEEE INTELLIGENT VEHICLES SYMPOSIUM
- 2 Paper in 2026 IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS

AI Innovation Park Internship Jul - Sep 2024

- Development of a time-series-based AI model for power prediction

Project

• Major Projects & Challenge

Preliminary Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Sep 2024 - Mar 2025

- Position : Team Member(model framework, dataset)
- RGB camera based End-to-End Mission-based Autonomous driving challenge in K-City

Final Round : 2025 Hyundai Motor Group's Autonomous Driving Challenge Apr 2025 - Oct 2025

- Position : Team Member(rule-based expert driving dataset, model framework)
- RGB camera based End-to-End Full-Self-Driving in K-City

2025 Brain to Society : Future mobility Jul - Dec 2025

- Position : Team Leader
- Project Title : Sensor-driven Hierarchical Information Fusion Transformer for BEV-based Maneuvering in Dense Multi-lane Environments

2026 Brain to Society : Future mobility Jun - Dec 2026

- Position : Team Leader
- Project Title : Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving

• Academic Projects

Low-cost Robot Manipulation (Robotics: Science and Systems) Dec 2024

- Position : Team Member
- Paper Title :Demonstrating a Vision-Based AI Robot for Strategic Board Games
- Project page: link

3D Vision and Machine Cognition Jun 2025

- Position : Team Member
- Project Title : Traffic-Light and Boundary-Aware BEV Learning from Multi-Camera Images

2D Vision: Human Pose Estimation Jun 2025

- Title : SoftEdge-PoseContrast View Conditioned Human-Pose Classifier with Contrastive Learning